scalar components q_{ai} and $\overline{q}_{ai}$ satisfy Eq. (27.4.11), which here reads

$$\sum_{abi} q^*_{ai}(t_A)_{ab}q_{bi} - \sum_{abi} \overline{q}^*_{ai}(t_A)_{ab}\overline{q}_{bi} = 0\,, \tag{29.3.25}$$

for all generators t_A of $SU(N_c)$. The gauge interactions (but not the superpotential) are invariant under simultaneous $SU(N_c)$ transformations on the color indices a of both q_{ai} and $\overline{q}_{ai}$; under independent $SU(N_f)$ and $\overline{SU(N_f)}$ transformations on the flavor indices i of q_{ai} and $\overline{q}_{ai}$, respectively, and under a $U(1)$ transformation of both q_{ai} and $\overline{q}_{ai}$ by opposite phases. Using these symmetries, it is possible to put the general solution of these conditions in the form

$$q_{ai} = \overline{q}_{ai} = \begin{cases} u_i\delta_{ai} & a \leq N_f \\ 0 & a > N_f \end{cases}\,, \tag{29.3.26}$$

where the u_i are complex numbers with the same phase. (Here is the proof. The $SU(N_c)$ generators t_A span the space of all traceless Hermitian matrices, so Eq. (29.3.25) is equivalent to the requirement that

$$\sum_i q^*_{ai}q_{bi} - \sum_i \overline{q}^*_{ai}\overline{q}_{bi} = k\delta_{ab}\,, \tag{29.3.27}$$

for some constant k. By a combined color and flavor transformation $q \to UqV$ with U and V unitary and unimodular, we can put the matrix q in the diagonal form (29.3.26), and by a unimodular change of phase of the diagonal elements we can arrange that they all have the same phase. Then Eq. (29.3.27) becomes

$$\sum_i \overline{q}^*_{ai}\overline{q}_{bi} = \begin{cases} \left(u_a^2 - k\right)\delta_{ab} & a \leq N_f \\ -k\delta_{ab} & a > N_f \end{cases}$$

The conditions for $a > N_f$ show that $k \leq 0$. If k were non-zero, then the $\overline{q}_{ai}$ would furnish N_c non-zero orthogonal vectors with N_f components, which is impossible for $N_f < N_c$, so $k = 0$. We can then put the $\overline{q}_{ai}$ in the diagonal form (29.3.26) by a series of unitary flavor transformations: first rotate $\overline{q}_{1i}$ into the 1-direction; then keeping the 1-direction fixed, rotate in the space perpendicular to this direction to put $\overline{q}_{2i}$ in the 2-direction; and so on; and then perform a unimodular phase transformation so that all diagonal elements have the same phase. Eq. (29.3.27) then shows that the absolute values of the diagonal elements of q_{ai} and $\overline{q}_{ai}$ are equal, and by a non-anomalous opposite phase change of q_{ai} and $\overline{q}_{ai}$ we can arrange that their common phases are equal, as was to be proved.)

The function H in Eq. (29.3.23) is here

$$H(q,\overline{q}) = \mathscr{I}\left[\mathrm{Det}_{ij}\sum_a q_{ai}\overline{q}_{aj}\right]^{-1/(N_c-N_f)} = \mathscr{I}\left[\prod_i u_i\right]^{-2/(N_c-N_f)}\,, \tag{29.3.28}$$